- I/O-intensive workloads access small amounts of data randomly. Provisioned IOPS SSD (io2) offers throughput guarantees, for example, for large and business-critical database workloads.
- Use Throughput Optimized HDD (st1) for workloads with sequential I/O and huge amounts of data, such as Big Data workloads. Don’t use this volume type for workloads in need of small and random I/O.
- Cold HDD (sc1) is a good fit when you are looking for a low-cost storage option for data you need to access infrequently and sequentially. Don’t use this volume type for workloads in need of small and random I/O.
- EBS Magnetic HDD (standard) is an older volume type from a previous generation. It might be a good option when you need to access your data infrequently.

Gibibyte (GiB) and Tebibyte (TiB)

The terms gibibyte (GiB) and tebibyte (TiB) aren’t used often; you’re probably more familiar with gigabyte and terabyte. But AWS uses them in some places. Here’s what they mean:

1 TiB = 240 bytes = 1,099,511,627,776 bytes	1 TB = 1012 bytes = 1,000,000,000,000 bytes
1 GiB = 230 bytes = 1,073,741,824 bytes	1 GB = 109 bytes = 1,000,000,000 bytes

Or, in other words, 1 TiB is 1.0995 TB and 1 GiB is 1.074 GB.

EBS volumes are charged based on the size of the volume, no matter how much data you store in the volume. If you provision a 100 GiB